Malachi D. Whitford, Ph.D.

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EDUCATION

Ph.D. in Ecology, San Diego State University & University of California, Davis **2015–2020**
Dissertation: Determinants of Predation Success: A Holistic Examination of the Predator-Prey Interactions Between Rattlesnakes and Kangaroo Rats

B.S. in Biology (Emphasis in Ecology and Biodiversity), Humboldt State University **2013–2015**
and Fresno City College **2011–2013**

PROFESSIONAL AND TEACHING EXPERIENCE

Environmental Science/Biology Professor (Full-time Tenured) **2021–Present**
Clovis Community College

- **Biol 17 – Wildlife and Plant Conservation Field Methods (4 units, 6 lab hours)**
  - Field-based course with an emphasis on field methods, species surveys, and ecological data collection.
- **Biol 7 – Field Biology (5 units, 6 lab hours)**
  - Field-based course with an emphasis on plant and vertebrate ID in the Central Valley and Sierra Nevada region.
- **Biol 2, 13 & 13L – Environmental Science (4 units, 3 lab hours)**
  - General overview of Earth's systems, with an emphasis on the biosphere.
- **Biol 3 – Introduction to Life Science (4 units, 3 lab hours)**
  - Introductory biology course for non-majors.
- **Biol 5 – Human Biology (4 units, 3 lab hours)**
  - Introductory human anatomy and physiology course**.**
- **Biol 11b – Biology for Majors 2 (5 units, 3 lab hours)**
  - Survey of major taxonomic groups.
- **PLS 1 & 1L – Plant Science (4 units, 3 lab hours)**
  - Introductory courses covering plant anatomy, physiology, taxonomy, ecology, and cultivation.
- **PLS 2 & 2L – Soil Science (4 units, 3 lab hours)**
  - **Introductory course covering the physical**, chemical, and biological components of the world's soils, with an emphasis on Central Valley soils.

Scientific Advisor, Life of Sound: Episode 1 – The Desert, Humblebee Films **2022**

- Provided expert advice and technical assistance in support of nature-themed documentary film projects.

Environmental Consultant (Staff Scientist and Project Manager) **2020–2021**
Colibri Ecological Consulting

- Primary job duties: rare/protected plant and animal surveys, water resource surveys, water resource impact assessments, soil sampling and surveys, environmental impact assessments, sensitive resource surveys, project and employee management, report/grant drafting, and construction monitoring.

Graduate Teaching Assistant **2015–2020**
San Diego State University

- **Bio 523 – Herpetology (4 units, 6 lab hours)**

  - Upper-division course on the physiology, anatomy, evolution, behavior, and ecology of reptiles and amphibians.
- **Bio 527 – Animal Behavior (3 units)**
  - Upper-division course covering topics in animal behavior and behavioral ecology.
- **Bio 215 & 291C – Biostatistics (3 units, 3 lab hours)**
  - Lower-division course covering statistical methods used in the biological sciences.
- **Bio 597 – Univariate Statistical Methods in Biology (3 units, 3 lab hours)**
  - Graduate-level course on univariate statistical techniques in biology.

Undergraduate Teaching Assistant: Field Biology (Biol 7) **Spring 2013**
Fresno City College

- Lower-division course on the natural history of local flora and fauna.

PEER-REVIEWED PUBLICATIONS (*indicates undergraduate co-author)

1. Hernandez, C.*, I. E. Visciglio*, F. Vang*, A. K. Serrano*, G. A. Freymiller, and **M. D. Whitford**. 2026. From Seeds to Scales: An Observation of a Kangaroo Rat (*Dipodomys deserti*) Scavenging a Desert Iguana (*Dipsosaurus dorsalis*). Western Wildlife 13:16–18.
2. Freymiller, G. A., **M. D. Whitford**, C. P. McGowan, T. E. Higham, and R. W. Clark. 2024. Springing into action: comparing escape responses between bipedal and quadrupedal rodents. Ecology and Evolution 14(9):e70292.
3. Iler, A. M., P. J. CaraDonna, L. K. Richardson, E. T. Wu, J. B. Fant, K. C. Pfeiler, G. A. Freymiller, K. N. Godfrey, A. J. Gorman, N. Wilson, **M. D. Whitford**, G. A. Edmonds, C. Stratton, and E. S. Jules. 2023. Genotype accounts for intraspecific variation in the timing and duration of multiple, sequential life-cycle events in a willow species. American Journal of Botany 110(2):e16112.
4. **Whitford, M. D.**, G. A. Freymiller, T. E. Higham, and R. W. Clark. 2022. Shaking things up: the unique feeding behaviour of western banded geckos when consuming scorpions. Biological Journal of the Linnean Society 135(3):533–540.
5. Freymiller, G. A., **M. D. Whitford**, M. J. Schwaner, C. P. McGowan, T. E. Higham, and R. W. Clark. 2022. Comparative analysis of *Dipodomys* species indicates that kangaroo rat hindlimb anatomy is adapted for rapid evasive leaping. Journal of Anatomy 240:466–474.
6. Robinson, K. E., M. L. Holding, **M. D. Whitford**, A. J. Saviola, J. R. Yates III, and R. W. Clark. 2021. Phenotypic and functional variation in venom and venom resistance of two sympatric rattlesnakes and their prey. Journal of Evolutionary Biology 34:1447–1465.
7. **Whitford, M. D.,** G. A. Freymiller, T. E. Higham, and R. W. Clark. 2020. The effects of temperature on the kinematics of rattlesnake predatory strikes in both captive and field environments. Integrative Organismal Biology 2(1):obaa025.
8. **Whitford, M. D.**, G. A. Freymiller, T. E. Higham, and R. W. Clark. 2020. The effects of temperature on the defensive strikes of rattlesnakes. Journal of Experimental Biology 223(14):jeb223859.

9. **Whitford, M. D.**, and A. P. Klimley. 2019. An overview of behavioral, physiological, and environmental sensors used in animal biotelemetry and biologging studies. Animal Biotelemetry 7:26.
10. **Whitford, M. D.**, G. A. Freymiller, T. E. Higham, and R. W. Clark. 2019. Determinants of predation success: How to survive an attack from a rattlesnake. Functional Ecology 33:1099–1109.
11. **Whitford, M. D.**, G. A. Freymiller, and R. W. Clark. 2019. Managing predators: the influence of kangaroo rat antipredator displays on sidewinder rattlesnake hunting behavior. Ethology 125:450–456.
12. Freymiller, G. A., **M. D. Whitford**, T. E. Higham, and R. W. Clark. 2019. Escape dynamics of free-ranging desert kangaroo rats (Rodentia: Heteromyidae) evading rattlesnake strikes. Biological Journal of the Linnean Society 127(1):164–172.
13. Freymiller, G. A., **M. D. Whitford**, T. E. Higham, and R. W. Clark. 2017. Recent interactions with snakes enhance escape performance of desert kangaroo rats (Rodentia: Heteromyidae) during simulated attacks. Biological Journal of the Linnean Society 122(3):651–660.
14. **Whitford, M. D.**, G. A. Freymiller, and R. W. Clark. 2017. Avoiding the serpent's tooth: predator-prey interactions between free-ranging sidewinder rattlesnakes and desert kangaroo rats. Animal Behaviour 130:73–78.
15. **Whitford, M. D.**, G. A. Freymiller, J. M. Ryan*, D. P. J. Steele*, C. N. Tapia*, and R. W. Clark. 2017. *Chionactis occipitalis*: hypomelanism. Herpetology Notes 10:411–412.
16. Higham, T. E., R. W. Clark, C. E. Collins, **M. D. Whitford**, and G. A. Freymiller. 2017. Rattlesnakes are extremely fast and variable when striking at kangaroo rats in nature: Three-dimensional high-speed kinematics at night. Scientific Reports 7:40412.
17. Clark, R. W., S. W. Dorr**, M. D. Whitford**, G. A. Freymiller, and B. J. Putman. 2016. Activity cycles and foraging behaviors of free-ranging sidewinder rattlesnakes (*Crotalus cerastes*): the ontogeny of hunting in a precocial vertebrate. Zoology 119(3):196–206.
18. Clark, R. W., S. W. Dorr, **M. D. Whitford**, G. A. Freymiller, and S. R. Hein. 2016. Comparison of anti-snake displays in the sympatric desert rodents *Xerospermophilus tereticaudus* (round-tailed ground squirrels) and *Dipodomys deserti* (desert kangaroo rats). Journal of Mammalogy 97(6):1709–1717.

ONGOING RESEARCH (*indicates undergraduate student)

1. Chandler, J.*, J. Brennan*, G. A. Freymiller, and **M. D. Whitford**. The habitat requirements of invasive squamates in California and potential areas of invasion.
2. Uy, V.*, G. Sprenger*, G. A. Freymiller, and **M. D. Whitford**. Predator deterrent signaling by coarse-haired pocket mice (*Chaetodipus* sp.).
3. Castillo, V.*, G. A. Freymiller, and **M. D. Whitford**. The impact of gray fox (*Urocyon cinereoargenteus*) digestion on the germination rate of coffeeberry (*Frangula californica*).

4. Hernandez, C.*, G. A. Freymiller, and **M. D. Whitford**. Species distribution modeling of California pipevine (*Aristolochia californica*) and the impacts of climate change on the southern extent of its range.

OTHER PUBLICATIONS

1. **Whitford, M. D.** 2026. OER Wildlife and Plant Conservation Field Methods Lab Manual. Clovis Community College.
2. **Whitford, M. D.** 2026. OER Soil Science Lab Manual. Clovis Community College.
3. **Whitford, M. D.** 2025. OER Plant Science Lab Manual. Clovis Community College.
4. **Whitford, M. D.**, G. A. Freymiller, and R. W. Clark. 2018. Rattlesnakes and Kangaroo Rats – Predator-Prey Interactions. Desert Report, Dec. 2018.
5. **Whitford, M. D.**, G. A. Freymiller, and R. W. Clark. 2018. Predator-Prey Interactions Between Rattlesnakes and Kangaroo Rats. Mojave National Preserve Newsletter, Sept. 2018.
6. **Whitford, M. D.** 2014. The Northern Pacific Rattlesnake. The Foothill Advocate 17(4):3.
7. Olsen, R., and **M. D. Whitford**. 2013. The Pacific Pond Turtle. The Foothill Advocate 17(1):4.

ACADEMIC, LEADERSHIP, AND INSTITUTIONAL SERVICE

**All activities completed at Clovis Community College unless otherwise stated.*

Curriculum Development and Oversight **2021–Present**

- Developed the Plant Science Certificate, Wildlife and Plant Technician Certificate, Plant AST, and Environmental Science AST.
- Created the curriculum for BIOL 17, BIOL 7, BIOL 13/13L, PLS 1/1L, PLS 2/2L.
- Full-time faculty lead for the Water Treatment and Distribution Program.

Program Review and Student Learning Outcome Assessment **2021–Present**

- Authored the 2-year program review reports for the Environmental Science Program and Water Treatment and Distribution Program.
- Assessed and authored the 2-year Student Learning Outcome Assessments and reporting for the Environmental Science Program and Water Treatment and Distribution Programs.

Committee Participation and Academic Leadership **2021–Present**

- Lead for the annual CTE Advisory Committees for the Environmental Science Program and Water Treatment and Distribution Program.
- Committee Membership:
    - Curriculum Committee, Natural & Health Science Representative 2023–Present
    - ZTC Workgroup 2022–Present
    - Career Technical Education Committee, Member 2021–2026
    - Academic Senate, Natural and Health Science Representative 2022–2023
    - Outcomes and Assessment Committee, Member 2021–2022
    - Faculty Search Committee Member for an AgTec Professor Spring 2026

  - Faculty Search Committee Member for a Biology Professor Spring 2025
  - Student Member for a Faculty Search Committee for a Biology Professor at San Diego State University Spring 2018
- Clovis Community College Conservation Club Advisor 2024–Present

Academic Service

- Ad Hoc Journal Reviewer:
- Behavioral Ecology and Sociobiology
- Communication Biology
- Diversity
- Frontiers in Behavioral Neuroscience
- Functional Ecology
- Herpetological Review
- Journal of Animal Ecology
- Journal of Evolutionary Biology
- Journal of Nature Conservation
- Journal of Zoology
- Oikos
- Scientific Reports
- Zoology

Academic Conference Volunteer Positions

- Student Presenter Mentor, Western Section of The Wildlife Society 2021
- Student Presentation Judge, Western Section of The Wildlife Society 2021
- Student Presentation Moderator, Southern California Animal Behavior Conference 2020

College and Community Service

- OER Reviewer for faculty-generated materials, including:
  - Field Biology Textbook and Wildlife and Plant Conservation Field Methods Textbook 2026
  - Biology for Majors 2 Textbook 2025
- Natural History Hike Docent, Sierra Foothill Conservancy 2012–2014 & 2022–Present
- Lead faculty for Clovis Community College field trip to Sequoia and Kings Canyon National Parks Spring 2026
- Lead faculty for Clovis Community College field trip to Morro Bay Spring 2025
- Lead faculty for Clovis Community College field trip to Pinnacles National Park Fall 2023
- Lead faculty for Clovis Community College field trip to Monterey Bay Fall 2022

GRANTS, SCHOLARSHIPS, AND HONORS

1. Whitford, M. D., G. A. Freymiller, and S. Coffman. 2025. Wildlife and Plant Conservation ZTC Pathway Grant, Academic Senate for California Community Colleges Open Educational Resources Initiative (ASCCC OERI), $25,000.

2. Whitford, M. D. 2024. Wildlife Identification and Conservation Field Techniques Project, State Center Community College Foundation Mini-Grant, $1,000.
3. Whitford, M. D. 2023. ZTC Field Biology: Materials to Remove Barriers for Students, State Center Community College Foundation Mini-Grant, $1,000.
4. Whitford, M. D. 2022. Educational Displays for Biology and Environmental Science, State Center Community College Foundation Mini-Grant, $900.
5. Whitford, M. D. 2021. Form and Function: Using Skulls of Local Species to Teach Adaptation and Species Identification, State Center Community College Foundation Mini-Grant, $773.
6. Whitford, M. D. 2019. Outstanding Teaching Assistant in Biology. San Diego State University.
7. Whitford, M. D. 2017. Naturally occurring predator-prey interactions: factors that influence performance and success in interactions between rattlesnakes and kangaroo rats, Joint Doctoral Program in Ecology funding, $1,500.
8. Whitford, M. D. 2017. Interpopulational variation in locomotor performance: how performance impacts predation success, Animal Behavior Society Student Research Grant, $2,000.
9. Whitford, M. D. 2016. Ecomechanics of a sit-and-wait predator, Joint Doctoral Program in Ecology funding, SDSU, $1,225.
10. Cum Laude, May 2015, Humboldt State University, Arcata, CA.

PRESENTATIONS, POSTERS, AND INVITED TALKS (**indicates undergraduate co-author***)**

1. Hernandez, C.*, I. E. Visciglio*, F. Vang*, A. K. Serrano*, G. A. Freymiller, and **M. D. Whitford**. 2026. From Seeds to Scale: An Observation of a Kangaroo Rat Scavenging a Desert Iguana. Clovis Community College Research Symposium, Fresno, CA.
2. Palafox, D.*, G. A. Freymiller, and **M. D. Whitford**. 2026. Benefits, Challenges, and Efficacy of Integrated Usage of PAMs and Machine Learning Birdcall Detection Algorithms in Avian Community Monitoring. Clovis Community College Research Symposium, Fresno, CA.
3. Castillo, M.*, G. A. Freymiller, and **M. D. Whitford**. 2026. Clovis Community College Field Guide, and a Visual Representation of the Importance of Scientific Illustration. Clovis Community College Research Symposium, Fresno, CA.
4. Ramirez, E.*, A. Serrano*, I. Visciglio*, D. Palafox*, and **M. D. Whitford**. 2026. Mediterranean House Gecko (*Hemidactylus turcicus*) Invasion Velocity Across California. Clovis Community College Research Symposium, Fresno, CA.
5. Selvans, S. M., D. Tinker, **M. Whitford**, K. Ford, M. Brady, B. Weinman, M. Richaud, A. H. Mine, and V. Walls. 2025. Shifting Student Understanding and Interest in Geoscience Careers Within Project Based 2YC Geoscience Courses Designed in Collaboration With Transfer Institution Partners. Geological Society of America Abstracts with Programs, Sacramento, CA.

6. **Whitford, M. D**. 2025. Lessons Learned from Group Projects. Fresno State NSF-Funded Professional Learning Community Summer Institute, Fresno, CA.
7. Hernandez, C.*, and **M. D. Whitford**. 2024. From Seeds to Scale: An Observation of a Kangaroo Rat Scavenging a Desert Iguana. Clovis Community College Research Symposium, Fresno, CA.
8. **Whitford, M. D.**, G. A. Freymiller, R. W. Clark, and T. E. Higham. 2020. The effects of temperature on the predatory and defensive strikes of rattlesnakes. Society for Integrative and Comparative Biology, Austin, TX.
9. **Whitford, M. D.**, G. A. Freymiller, R. W. Clark, and T. E. Higham. 2019. The effects of temperature and body size on strike performance in rattlesnakes. Biology of Pit Vipers, Rodeo, NM.
10. **Whitford, M. D.**, G. A. Freymiller, R. W. Clark, and T. E. Higham. 2017. Three-dimensional kinematics of rattlesnake strikes in nature. Society for Integrative and Comparative Biology, New Orleans, LA.
11. **Whitford, M. D.**, R. W. Clark, S. Dorr, and G. Freymiller. 2016. Avoiding the serpent's tooth: predator-prey interactions between free-ranging sidewinder rattlesnakes and desert kangaroo rats. Southwestern Organismal Biology Conference, Fullerton, CA.
12. **Whitford, M. D.**, R. W. Clark, S. Dorr, and G. Freymiller. 2016. Avoiding the serpent's tooth: predator-prey interactions between free-ranging sidewinder rattlesnakes and desert kangaroo rats (poster). San Diego State Student Research Symposium, San Diego, CA.
13. **Whitford, M. D.**, R. W. Clark, S. Dorr, and G. Freymiller. 2015. Response of free-ranging sidewinder rattlesnakes (*Crotalus cerastes*) to the anti-snake displays of the desert kangaroo rat (*Dipodomys deserti*) (poster). College of Natural Resources-Louis Stokes Alliances for Minority Participation Student Research Symposium, Arcata, CA.

MENTORED STUDENTS

- James Brennan, Clovis Community College Undergraduate 2026–Present
- Hilario Cortez, Clovis Community College Undergraduate 2026–Present
- Vincy Uy, Clovis Community College Undergraduate 2026–Present
- Gavin Sprenger, Clovis Community College Undergraduate 2026–Present
- Valentine Castillo, Clovis Community College Undergraduate 2026–Present
- Jheta Chandler, Clovis Community College Undergraduate 2025–Present
- Dylan Palafox, Clovis Community College Undergraduate 2025–Present
- Chris Hernandez, Clovis Community College Undergraduate 2023–Present
- Alyssa Serrano, Clovis Community College Undergraduate 2025–2026
- Isa Visciglio, Clovis Community College Undergraduate 2025–2026
- Fiona Vang, Clovis Community College Undergraduate 2025–2026
- Elizabeth Ramirez, Clovis Community College Undergraduate 2024–2026
- Haiden Del Fierro, Clovis Community College Undergraduate 2023–2025

SELECT MEDIA COVERAGE, INTERVIEWS, AND PODCASTS

Media Coverage: Whitford et al. 2019

1. **PBS Deep Look**. "Kangaroo Rats Are Furry, Spring Loaded Ninjas." March 2020
2. **BBC Outlook**. "The 'Ninja rats' taking on snakes and winning." July 2019
3. **Daily Aztec**. "SDSU researcher gains insight into the defense mechanisms used by kangaroo rats." April 2019
4. **Science News**. "Watch a desert kangaroo rat drop kick a rattlesnake." March 2019
5. **Washington Post**. "'Ninja' kangaroo rats kick rattlesnakes in the head faster than you can blink." March 2019

Media Coverage: Whitford et al. 2017

1. **New York Times: Science Take**. "In Rat vs. Snake, an Unlikely Victor." November 2017
2. **National Geographic**. "Watch a rat escape a snake attack in 100 milliseconds." July 2017
3. **Global News Canada**. "Slow motion footage shows kangaroo rat narrowly escaping rattlesnake's deadly bite using 'surprising' techniques." July 2017
4. **The Society for Integrative and Comparative Biology**. "Bioscience Talks." July 2017
5. **Center for Biological Diversity**. "Flips, face kicks, sand throwing." July 2017

SELECT PROFESSIONAL DEVELOPMENT

1. From Data to Deep Time Visual Narratives, Clovis Community College 2026
2. Navigating Faculty Evaluations, Clovis Community College 2026
3. Beyond Open Educational Resources and Zero Textbook Cost Materials, Clovis Community College 2026
4. SLO Instrumentation and Process Review, Clovis Community College 2025
5. Biology Program Data Evaluation via Tableau, Clovis Community College 2025
6. AI Policies and Embedded Tutor Best Practices, Clovis Community College 2025
7. Curriculum in CurriQunet, Clovis Community College 2024
8. Crush Climate Corps: Fostering Student Innovation, Clovis Community College 2024
9. Biology SLO Assessments and Equipment Training, Clovis Community College 2024
10. Fresno State NSF-Funded Professional Learning Summer Institute 2023
11. Online Teaching Certification, Clovis Community College 2023
12. Labster Workshop, Clovis Community College 2023
13. Unit Program Planning, Clovis Community College 2023
14. Panopto and Anatomage Table Integration, Clovis Community College 2023
15. Ungrading for Learning and Equity, California Virtual Campus 2022
16. Increasing Retention of Underrepresented Groups in STEM, Clovis Community College 2022
17. Equitable Grading Practices, Clovis Community College 2022
18. Universal Design for Learning (UDL), Clovis Community College 2021
19. On Course Workshop, Clovis Community College 2021
20. Finding OER and Customizing Canvas, Clovis Community College 2021

21. IRACDA: Creating and Implementing an Engaging, Equitable, Inclusive, and Community-Oriented Virtual Classroom, San Diego State University 2020
22. Learning to Teach Online, University of New South Wales 2020
23. Certificate in Evidence-Based Teaching, San Diego State University 2019
24. IRACDA: Scientific Teaching Workshop (Recognizing and facilitating inclusivity in the classroom), San Diego State University 2019